MIL-T-37047A (DSA-DM)
3 October 1974
SUPERSEDING
MIL-T-37047 (DSA-DM)
9 April 1974

MILITARY SPECIFICATION

# TEST TUBE, 17 by 100 mm., Polypropylene, Disposable, with Snap-on Cap

This limited coordination Military specification has been prepared by the Defense Personnel Support Center, based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of MIL-T-37047 (DSA-DM). However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

## 1. SCOPE

1.1 This specification covers a sterile, disposable test tube with snap-on cap suitable for use in laboratory culture procedures.

## 2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-T-76 | Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing). |
| MMM-A-250 | Adhesive, Water-Resistant (for Sealing Waterproofed Paper). |
| DD-G-541 | Glass, Laboratory. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Shipping, Fiberboard. |

Military

| | |
|---|---|
| MIL-S-36586 | Sterilization Test Strip Set, Bacterial Spore. |
| MIL-L-10547 | Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible. |



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STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. The tube and cap shall be fabricated of translucent polypropylene plastic.

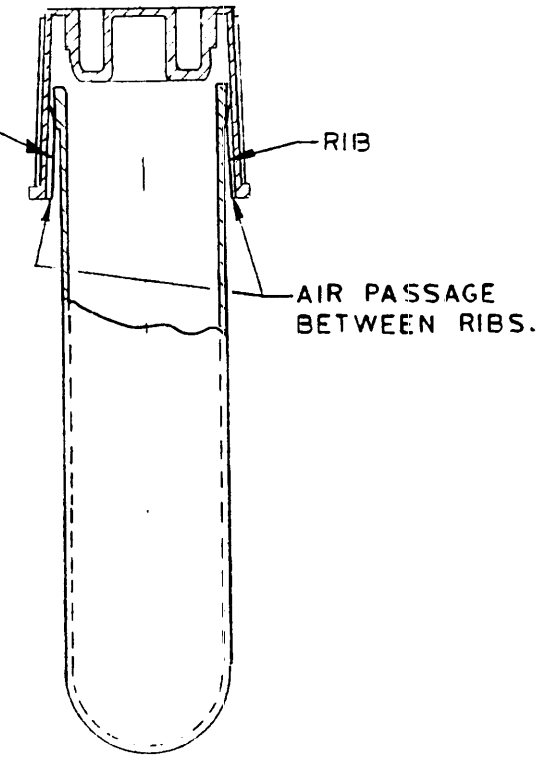
* 3.2 Construction. For purposes of identification, a test tube shall be a tube and snap-on cap enclosed in a plastic bag. Plastic bag shall maintain sterility of its contents unless damaged or opened. Tube and cap shall be so designed that they form a friction fit in two positions as shown in Figure 1. Test tube shall be furnished with the tube assembly in Position 1 which shall seal loosely and allow air to pass into the tube for aerobic culturing. Position 2 shall seat securely to form a liquid-tight seal.

3.3 Capacity. Overflow capacity of tube shall be 16 $\pm$ 0.5 milliliters (ml.) when tested as specified in 4.4.1.

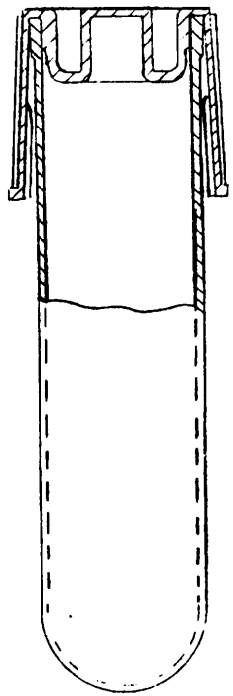
* 3.4 Dimensions. Outside diameter of tube shall be 17 $\pm$ 0.5 millimeters (mm.) Nominal length of tube shall be 100 mm. Length at Position 1 of Figure 1 shall be less than 110 mm.; length at Position 2 shall be more than 90 mm.



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POSITION 1



POSITION 2

FIGURE 1

NOTE:

DESIGN DETAILS OF TUBE AND CAP ARE FOR ILLUSTRATION ONLY. ALTERNATE DESIGNS MAY BE USED PROVIDING THE UNIT MEETS THE REQUIREMENTS OF THE SPECIFICATION.

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* 3.5 Toxicity. Tube shall not be toxic when tested as specified in 4.4.2.

* 3.6 Leakage. Tube and cap shall not leak when tested as specified in 4.4.4.

3.7 Sterility. The tube and cap shall be sterile when tested with a biological indicator as specified in 4.4.3.

3.7.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National Stock Number (NSN).
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.7.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).



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3.8 Workmanship. Test tube shall be free of defects listed in Table III and other defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105.

Table I Sampling for examination

| Unit of product | Category | Inspection level | AQL (percent defective) |
|---|---|---|---|
| For visual examination | | | |
| Tube | Major A | S-4 | 1.0 |
| Tube and cap | Major B | S-1 | 1.0 |
| Tube | Minor | S-4 | 2.5 |
| For dimensional examination | | | |
| Tube and cap | --- | S-1 | 2.5 |



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4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table II and MIL-STD-105. Unit of product shall be one package, and acceptance number shall be zero, except for the leakage test.

* Table II Sampling for tests

| Characteristic | Requirement | Test procedure | Inspection level | AQL (percent defective) |
|---|---|---|---|---|
| Capacity | 3.3 | 4.4.1 | S-1 | |
| Toxicity | 3.5 | 4.4.2 | 18 tubes and caps | |
| Leakage | 3.6 | 4.4.4 | S-3 | 4.0 |
| Sterility | 3.7 | 4.4.3 | S-3 | |

4.3 Examination. The test tube shall be examined to determine compliance with all requirements contained in this specification. Examination shall include, but shall not be limited to, the defects listed in Table III.

Table III Classification of defects

| Category | Defects |
|---|---|
| Major A | |
| 101 | Bag not free of tears and holes. |
| 102 | Tube or cap not free of oil, water drops, dirt and other foreign material. |
| 103 | Tube or cap not free of cracks. |
| Major B | |
| 151 | Gate stub not free of sharpness or extends more than 1/64-inch. |
| 152 | Tube or cap not free of flash which exceeds 1/64-inch. |
| 153 | Tube or cap does not seat securely as shown in Position 2 of figure. |
| 154 | Inside of tube not smooth, other than gate stub area (see 151 and 201). |
| Minor | |
| 201 | Tube or cap not free of scuffs and die marks other than gate stub at bottom of tube (see 151). |

4.3.1 Dimensional examination. The tube and cap shall be examined for defects in dimensions. Any dimensions not within the tolerances specified herein shall be classified as a defect.



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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Capacity. Capacity to the top of the tube, overflow capacity, shall be determined gravimetrically or volumetrically.

* 4.4.2 Toxicity. Tubes and media necessary for tests are:

a. Control tubes. Tubes and caps shall be glass or polypropylene. Glass shall be in accordance with DD-G-541, class A or B. Polypropylene tubes and caps shall be from a previous lot, which has been approved and delivered to the Government; use of tubes must be approved by the contracting officer.

b. Sample tubes shall be as indicated in 4.2.2.

c. Soybean casein digest broth U.S.P.

d. 24-hour soybean casein digest broth cultures of the following organisms:
(1) Bacillus subtilis (B. subtilis)
(2) Streptococcus pyogenes (S. pyogenes) Group A, ATCC 12384 and 10526
(3) Sarcina lutea (S. lutea) ATCC 9341

e. Seed agar

Inoculate 2 petri dishes with 1 ml. of $1 \times 10^{-7}$ dilution of B. subtilis. Repeat inoculation of additional pairs of petri dishes with $1 \times 10^{-6}$ and $1 \times 10^{-5}$ dilutions of B. subtilis and with $1 \times 10^{-7}$, $1 \times 10^{-6}$ and $1 \times 10^{-5}$ dilutions of S. pyogenes and S. lutea. Pour 10 to 20 ml. of seed agar into each petri dish. Swirl dishes sufficiently to disperse inoculum evenly. Incubate for not less than 48 hours or more than 72 hours at 30-35° Centigrade (°C.). Count and record the number of colonies in each petri dish. Observation of growth in petri dishes is for purposes of determining suitability of culture dilutions for tube testing. To be suitable for application to tube testing, dilutions must produce between 10 and 100 colonies in a petri dish. Disregard tube results for dilutions which do not produce between 10 and 100 colonies.

Put 5 ml. of soybean casein digest broth in each sample and in each control tube. Close caps to Position 2. Incubate tubes at 30-35°C. for 3 days. Inoculate 2 sample tubes and 2 control tubes with 1 ml. of $1 \times 10^{-7}$ dilution of B. subtilis. Repeat inoculation of additional pairs of tubes with $10^{-6}$ and $10^{-5}$ dilutions of B. subtilis and with $1 \times 10^{-7}$, $1 \times 10^{-6}$ and $1 \times 10^{-5}$ dilutions of S. pyogenes and S. lutea. Incubate all tubes at 30-35°C. for not less than 48 hours or more than 72 hours. Observe applicable tubes for bacterial growth.



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Tube shall be considered non-toxic if there is growth in both the sample and control tubes for applicable dilutions. Tube shall be considered toxic if there is growth in the control tube but no growth in the sample tube for applicable dilutions.

4.4.3 Sterility test. The tube and cap shall be tested for sterility in accordance with the method specified in the U.S.P. for Procedural Details for Sterility Tests of Class B, Type II Inoculated Product or Inoculated Carrier for Parenteral Devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.3.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II.

4.4.3.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

| Type of Sterilization | Organism | Incubation Temperature |
|---|---|---|
| Steam | B. stearothermophilus | 55°- 65°C. |
| Dry heat | B. stearothermophilus or | 55°- 65°C. |
| | B. subtilis var.niger (or globigii) | 30°- 35°C. |
| Gas | B. subtilis var. niger (or globigii) | 30° - 35°C. |
| Radiation | B. pumilus | 30°- 35°C. |

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.3.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.3 and 4.4.3.1. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.3.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P.

4.4.3.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests



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for approval of any alternate method of testing for sterility shall be made to the contracting officer, and approval obtained before employment of the alternate method on a government contract.

4.4.3.4 Product testing. Testing for sterility of the product after receipt by the Government shall be without a biological indicator and in accordance with the U.S.P.

* 4.4.4 Leakage. Fill tube to within 4 mm. of the top (line formed by the bottom of the protrusion shown on Figure 1 is approximately 4 mm. from the top) with an aqueous dilution of dye. Put cap in Position 2. Put toweling on the bottom of a vacuum chamber. Insert inverted tube assembly in vacuum chamber and pull a 10-inch vacuum for 5 minutes. Turn off vacuum and allow chamber to come to atmospheric pressure. Check for evidence of leakage. Dye or water on the portion of the cap and tube past the seal or on the toweling shall be considered evidence of leakage.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing 500 test tubes, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | Not required | 2 units |

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

* 5.1.3.1 Unit package. Five hundred (500) test tubes, individually packaged as specified in 3.2, shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall



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be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

* 5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. In addition, contract or purchase order number and name of contractor shall be shown. Marking shall include the word "Sterile" and a legend indicating that sterility is not guaranteed after inner bag or envelope seal has been broken. Lot (control) number shall be shown. In addition, the method of sterilization (G-Gas, R-Radiation, C-Cold,



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S-Steam, D-Dry heat sterilization) shall be shown following the level of packaging and the date of packaging. EXAMPLE: "A-5/71-G". NOTE: Final sterilization shall be on or after indicated packaging date.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, marking shall include the method of sterilization markings specified in 5.3.1.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items as listed in the Federal Supply Catalog:

| National Stock No. | Item Identification |
|---|---|
| * 6640-00-759-2152 | TEST TUBE, 17 by 100 mm., Polypropylene, Translucent, Disposable, with Snap-on Cap, 500s |

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet Military requirements.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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Review activities:

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Project No. 6640-0909

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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